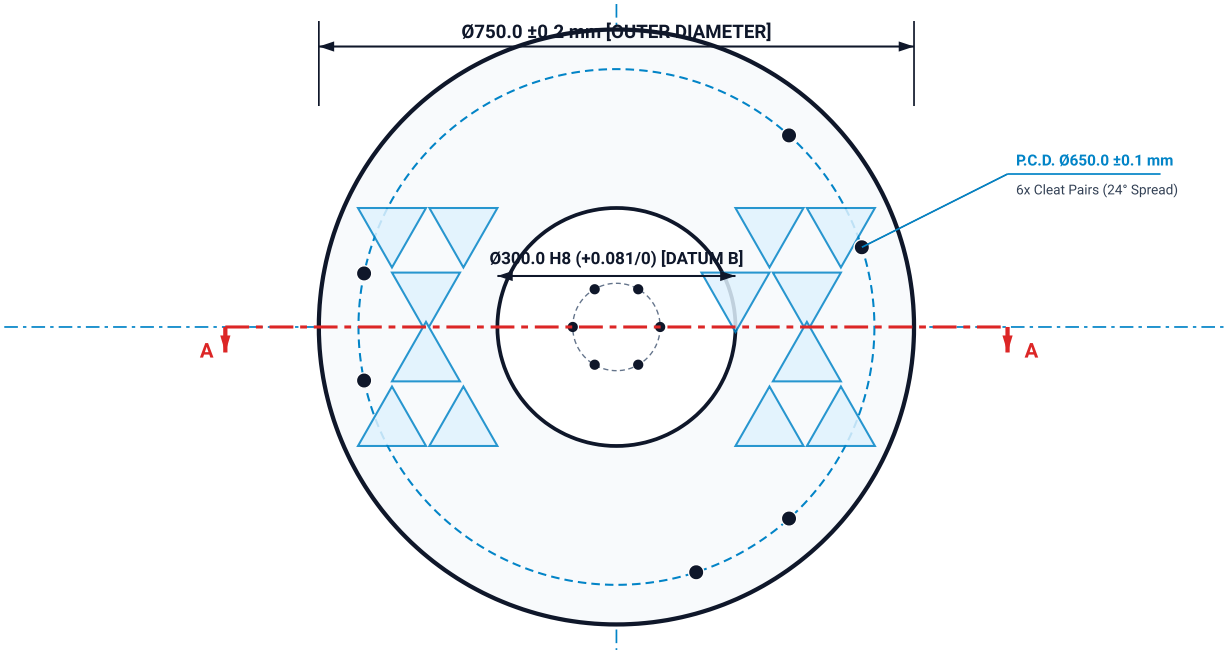
TOP BASE ISOGRID RING PLATE • 2D ORTHOGRAPHIC VIEWS

3RD ANGLE PROJECTION

Part No: MGSE-TOP-001 | Material: Aluminum 7075-T6 Billet | Mass: 10.02 kg



SECTION A-A • TRUE-SCALE ISOGRID WALL & FLOOR CUT (Scale 1:1)



GENERAL AEROSPACE SPECIFICATIONS & MANUFACTURING INSTRUCTIONS:

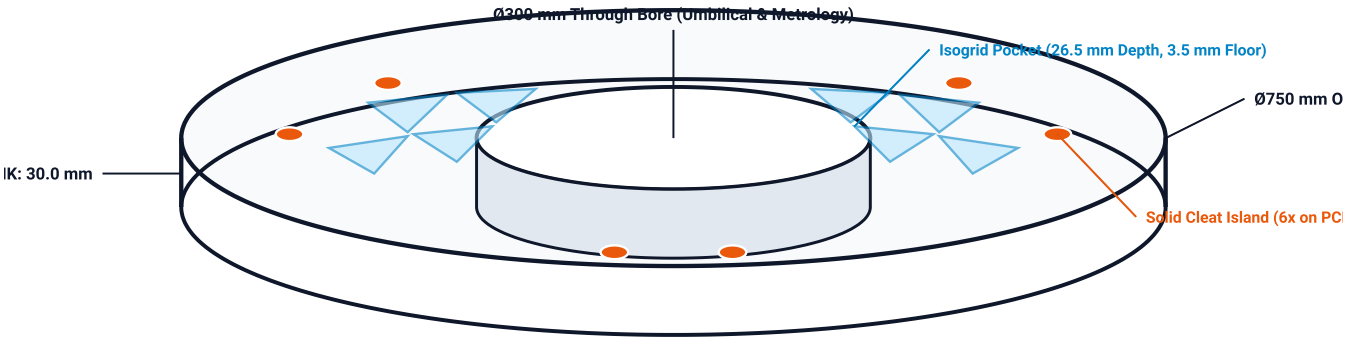
1. **MATERIAL:** Aluminum Alloy 7075-T6 forged billet per AMS 4045 / ASTM B209. Ultrasonic NDT inspected Class AA prior to CNC milling.
2. **GEOMETRIC STRESS RELIEF:** Rough pocketing leaving +1.5 mm stock → thermal cycle stabilization (-40°C to +100°C) → finish skim cuts to hold flatness.
3. **SURFACE FINISH:** Datum A top face Ra ≤ 0.8 µm; center bore Datum B Ra ≤ 0.4 µm; all pocket floors & ribs Ra ≤ 1.6 µm. Internal corner fillets R3.0 mm mandatory.
4. **SURFACE TREATMENT:** Clear Hard Anodize per MIL-A-8625, Type III Class 1 (non-dyed, 50 µm thickness). Sealed in cleanroom nickel acetate (CVCM < 0.10%).
5. **CLEAT MOUNTING:** 24x M8x1.25 tapped holes (depth 18 mm min) & 12x Ø6.0 m6 reamed dowel holes. Proof tested to 2.0x WLL = 2.45 kN.

ICEYE SATELLITE MGSE HEXALIFT 6-DOF GROUND SUPPORT	**PART NAME**	**TOP BASE ISOGRID RING PLATE**		**DRAWING NO.**	**DWG-MGSE-TOP-001**
	STANDARDS	ASME Y14.5-2018	**MATERIAL**	Al 7075-T6 Billet	**REVISION**	**REV D (FINAL)**
SCALE: 1:5 [A4 METRIC]	**WEIGHT:** 10.02 kg	**GD&T:** Flatness &Empty; 0.05 A		**TRUE POSITION:** ⌖ Ø0.08Ⓜ A B	

TOP BASE ISOGRID RING PLATE • 3D ISOMETRIC PROJECTION & HOLE TABLE

3D VIEW

Part No: MGSE-TOP-001 | Sheet 2: Trimetric View & Precision Fastener Coordinates



FASTENER & DOWEL COORDINATE SCHEDULE (POLAR & CARTESIAN FROM CENTER ORIGIN 0,0)

HOLE ID	QTY	NOMINAL HOLE SPECIFICATION	PCD / BCD	ANGULAR POSITIONS	DEPTH	TOLERANCE / FIT
H1 – H24	24	M8 × 1.25 – 6H Threaded Mounting Holes	**Ø650.0 mm**	48°, 72°, 168°, 192°, 288°, 312° (±tangent)	18 mm min	Pos Ø0.08Ⓜ A B
D1 – D12	12	Ø6.0 mm Reamed Dowel Shear Holes	**Ø650.0 mm**	Cleat Centerlines (2 per station)	15 mm	Ø6.0 m6 (+0.012/+0.004)
L1 – L6	6	M10 × 1.5 – 6H Crane Hoist Bail Retention	**Ø110.0 mm**	60°, 120°, 180°, 240°, 300°, 360°	22 mm min	Pos Ø0.10Ⓜ A B

ICEYE SATELLITE MGSE HEXALIFT 6-DOF GROUND SUPPORT	**PART NAME**	**TOP BASE ISOGRID RING (3D & HOLE SCHEDULE)**		**DRAWING NO.**	**DWG-MGSE-TOP-001**	
	STANDARDS	ASME Y14.5-2018	**MATERIAL**	Al 7075-T6 Billet	**REVISION**	**REV D (FINAL)**
PROJECTION: 3D Trimetric	**UNITS:** mm [Degrees]	**APPROVAL:** Lead Smart Manufacturing & Structural Verification Engineer				